

MELUN VILLAROCHE AP, France

WMO: 071530

Lat: 48.6106N

Lon: 2.6794E

Elev: 92

StdP: 100.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.2	(c) -3.3	(d) -8.6	(e) 1.8	(f) -3.6	(g) -6.3	(h) 2.2	(i) -2.2	(j) 13.7	(k) 10.6	(l) 12.4	(m) 9.9	(n) 2.5	(o) 0	(p) 0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.3	(c) 31.2	(d) 20.5	(e) 29.0	(f) 19.7	(g) 27.1	(h) 19.0	(i) 21.6	(j) 28.9	(k) 20.6	(l) 27.2	(m) 19.7	(n) 25.6	(o) 3.6	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.1	24.3	18.3	13.3	22.9	17.5	12.6	22.2	63.2	29.0	59.5	27.2	56.4	25.7	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	9.4	8.3	-7.9	35.2	3.2	2.4	-10.2	37.0	-12.1	38.4	-13.9	39.8	-16.2	41.5
			-8.3	23.6	3.2	1.0	-10.6	24.3	-12.4	24.9	-14.2	25.4	-16.5	26.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	4.0	4.8	7.7	10.5	14.1	17.4	19.5	19.1	15.7	12.2	7.4	4.7	(6)
(7)		DBStd	6.49	4.11	3.70	3.32	3.42	3.28	3.30	3.16	3.13	2.89	3.46	3.58	3.87	(7)
(8)		HDD10.0	737	187	146	86	33	4	0	0	0	1	18	93	169	(8)
(9)		HDD18.3	2660	444	378	329	235	137	56	23	26	89	192	329	423	(9)
(10)		CDD10.0	1274	2	2	15	49	132	223	294	283	171	86	14	3	(10)
(11)		CDD18.3	156	0	0	0	1	6	28	59	51	9	1	0	0	(11)
(12)		CDH23.3	1629	0	0	0	15	71	298	617	527	93	8	0	0	(12)
(13)		CDH26.7	516	0	0	0	1	9	86	209	193	18	0	0	0	(13)
(14)	Wind	WSAvg	3.8	4.4	4.4	4.2	3.9	3.5	3.3	3.4	3.3	3.3	3.7	3.9	4.4	(14)
(15)	Precipitation	PrecAvg	663	52	48	43	51	65	58	62	57	47	60	57	62	(15)
(16)		PrecMax	903	114	92	115	118	157	121	152	103	98	132	110	149	(16)
(17)		PrecMin	509	5	7	7	6	6	7	20	7	0	20	15	18	(17)
(18)		PrecStd	100	23	24	26	30	30	29	33	27	27	26	25	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	15.8	20.3	25.3	27.9	32.4	34.4	35.2	29.0	24.4	17.6	14.0	(19)
(20)			MCWB	11.9	10.8	12.5	16.7	19.6	21.9	21.3	21.7	18.8	17.5	14.6	12.4	(20)
(21)		2%	DB	12.2	12.7	17.6	22.4	25.3	29.0	31.1	31.3	26.1	21.4	15.5	12.7	(21)
(22)			MCWB	10.9	9.8	11.9	14.5	17.7	20.5	20.5	20.1	18.2	16.6	13.5	11.4	(22)
(23)		5%	DB	11.0	11.3	15.2	19.5	23.1	26.7	28.9	28.7	23.6	19.1	14.1	11.6	(23)
(24)			MCWB	9.6	9.4	11.0	13.2	16.6	19.4	19.7	19.4	17.2	15.5	12.4	10.5	(24)
(25)		10%	DB	9.6	10.2	13.3	17.1	20.8	24.4	26.9	26.2	21.6	17.3	12.6	10.4	(25)
(26)			MCWB	8.4	8.6	10.1	11.9	15.4	18.1	18.9	18.5	16.2	14.5	11.3	9.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.4	11.8	13.8	17.2	20.6	23.1	22.6	22.5	20.2	18.5	15.2	12.9	(27)
(28)			MADB	13.3	13.7	18.0	23.9	26.5	30.3	31.2	31.5	26.2	22.5	16.6	13.7	(28)
(29)		2%	WB	11.1	10.5	12.6	15.2	18.6	21.3	21.3	21.3	19.0	17.1	13.8	11.7	(29)
(30)			MADB	12.1	12.0	16.0	20.9	23.9	27.3	29.0	28.8	24.1	20.4	15.4	12.5	(30)
(31)		5%	WB	9.7	9.7	11.6	13.7	17.2	20.0	20.4	20.2	17.9	16.0	12.6	10.6	(31)
(32)			MADB	10.8	11.1	14.3	18.4	21.6	25.5	27.2	26.7	22.4	18.6	13.9	11.5	(32)
(33)		10%	WB	8.6	8.7	10.5	12.5	15.9	18.7	19.4	19.2	17.0	15.0	11.5	9.4	(33)
(34)			MADB	9.5	10.1	13.0	16.3	20.1	23.4	25.4	24.7	20.7	17.0	12.6	10.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	6.7	8.8	10.4	10.2	10.8	11.3	11.4	10.5	8.4	6.0	5.1	(35)
(36)			MADB	6.1	8.4	12.3	14.3	13.9	14.5	15.4	15.8	13.8	10.6	7.4	5.8	(36)
(37)			MCWBR	5.2	6.1	7.5	7.7	7.3	7.3	6.1	6.4	6.9	6.3	5.5	4.9	(37)
(38)		5% WB	MADB	5.9	7.6	10.2	12.7	12.6	13.4	13.3	13.7	11.6	9.5	6.8	5.6	(38)
(39)			MCWBR	5.2	5.9	6.7	7.2	7.1	7.3	5.9	6.3	6.3	6.1	5.4	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.331	0.352	0.386	0.411	0.417	0.421	0.420	0.409	0.379	0.371	0.343	0.327	(40)	
(41)		taud	2.398	2.349	2.251	2.196	2.220	2.245	2.244	2.297	2.363	2.376	2.403	2.371	(41)	
(42)		Ebn at Noon	724	791	819	836	845	843	837	830	819	757	701	671	(42)	
(43)		Edh at Noon	68	89	114	133	135	133	131	119	101	85	67	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.92	1.71	2.91	4.38	5.17	5.77	5.45	4.76	3.66	2.18	1.11	0.77	(44)	
(45)		RadStd	0.11	0.25	0.36	0.49	0.52	0.52	0.46	0.47	0.33	0.17	0.11	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page